MILITARY SPECIFICATION

ACCELEROMETER GROUP, COUNTING
TYPE DESIGNATION MS 25447 AND MS 25448

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the design and performance requirements for a unidirectional, four-nominal load-factor level counting accelerometer group.

1.2 Classification - The Counting Accelerometer Group covered by this specification shall consist of one each of the following:

Item	Type Designation
Transducer	MS 25447 (1 thru 10)
Indicator	MS 25448

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-P-666	Primer Coating, Zinc Yellow, for Aluminum and Magnesium Surfaces

Military

MIL-W-5088	Wiring; Aircraft, Installation of

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MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-E-15090	Enamel Equipment, Light Grey
MIL-C-15328	Coating, Pretreatment
MIL-T-18303	Test Procedures; Preproduction and For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
AR-8	Versatile Avionic Shop Test System, Avionic System Compatibility, General Requirements for
AR-9	Vast Test Programs, General Requirements for
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for

STANDARDS

Federal

FED-STD-595	Colors

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Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS25447	Transducer, Counting Accelerometer
MS25448	Indicator, Counting Accelerometer
MS33558	Pointers, Dials, Standard Design of Aircraft Instrument

2.1.1 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.6)

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(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of Specification AR-5.

3.2.3 Modules, Maintenance - The electronic portions of the equipment shall be divided into maintenance modules (see Specification AR-10). Maintenance modules shall normally be considered nonrepairable.

3.2.4 Nonmagnetic Materials - Nonmagnetic materials shall be used for all parts of the counting accelerometer except where magnetic materials are essential.

3.2.5 Filling Medium - When required by the transducer design, a gas or oil may be used which conforms with the equipment characteristics and requirements of this specification.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 5 pounds.

3.3.2 Reliability -

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for at least 500 operations without the necessity for readjustment of controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 100,000 operations with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

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3.3.2.3 Reliability in Mean Time Between Failures (MTBF) - For the test program under 4.4.3, the mean (operating) time between failures shall be 50,000 operations.

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400 and Figure 1 of this specification.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Interchangeability of Reordered Equipment - For reordered equipment, interchangeability shall exist between all units and all replaceable assemblies and subassemblies of any previously manufactured equipment as supplied or designated by the procuring activity. Such interchangeability shall be measured against the supplied or designated model, drawings, or other technical information provided for this purpose. The contract or order shall stipulate whether the model, drawings, or other information govern.

3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.3.6 Interface Requirements - The equipment shall be capable of satisfactory operation while using the external wiring conforming with the detailed requirement of Figure 1. The equipment shall be interchangeable with all units previously procured by this specification. Internal and external wiring, dimensions, design, assemblies, circuits, etc., shall conform with the required interface of all equipment of previous procurements.

3.3.7 Provisions for Maintainability - Provisions for maintenance shall be as specified in MIL-E-5400. Built-in test equipment, construction and packaging, and provisions for test points to the greatest extent practicable for ease of field testing and maintenance shall be as specified in Specification AR-10. Equipments which are required to be compatible with the Versatile Automatic Shop Test System (VAST) shall be in accordance with Specification AR-8. When required, VAST Test Programs shall be furnished in accordance with Specification AR-9.

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3.3.8 Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	27.5 ± 0.5 V DC

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.3.11 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Figure 2.

3.3.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) DC power, +28V Category A, 1.5 Amps

3.3.13 Test Conditions and Equipments -

3.3.13.1 Standard Conditions - Unless otherwise specified, all tests required by this specification shall be made under the conditions of 3.3.9 herein.

3.3.13.2 Attitude - Unless otherwise specified, the accelerometer group shall be tested with the sensing axis normal to the direction of the earth's gravitational field.

3.3.13.3 Vibration Stand - The vibration stand shall be capable of subjecting the accelerometer groups to accelerations from 0.1 to 10.5g within the frequency range of 5 to 2000 cps.

3.3.13.4 Static Calibration and Dynamic Test Equipment - The equipment used for static calibration and dynamic tests shall be capable of determining load-factor levels within the limits required by paragraphs 4.4.1.4 and 4.4.2.4 of this specification.

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3.4 Detail Requirements -

3.4.1 Counting Accelerometer Group - The equipment shall be utilized to count aircraft maneuver-load factors. The counting accelerometer group shall consist of two components: (a) A unidirectional acceleration sensing device, referred to as the transducer, which shall be capable of completing electrical circuits at discrete load-factor levels and shall be hermetically sealed, and (b) a counting accelerometer indicator, referred to as the indicator, which shall have counter displays to indicate the number of times the aircraft is subjected to load factors of preset levels. The group, when electrically interconnected shall provide signals and control circuits as shown in Figure 1.

3.4.1.1 Wiring - The internal wiring of the accelerometer group shall be in accordance with MIL-E-5400 and shall operate satisfactorily in the environment range stated herein. The wiring of each component (transducer/indicator) shall comply with the requirements specified and Figure 1 of this specification.

3.4.1.2 Under-Voltage Protection - The equipment shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

3.4.1.3 Painting - External surfaces shall be painted in accordance with MIL-E-15090, formula II using FED-STD-595, color 26329.

3.4.1.4 The protective coating and finishing procedure shall conform with the requirements of MIL-C-15328 and FED-TT-P-666.

3.4.1.5 Case - The components shall be constructed of lightweight, nonmagnetic metal, uniform in texture and having a smooth surface.

3.4.1.6 Dynamic Response - The dynamic response of the counting accelerometer group to sinusoidal inputs shall be flat to 2.5 cps. The equipment shall indicate counts at single-amplitude, nominal load-factor levels prescribed in Table 1, within the following limits:

(1) $\pm$ 0.20g for load-factor levels of 2.0g to 7.0g.

(2) $\pm$ 0.30g for load-factor levels of 7.5g to 10.0g.

From 3 cps, the response shall drop at a minimum rate of 12 db per octave. The indicator shall not indicate a count or operation with sinusoidal or square wave inputs of 4.0 cps and above.

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3.4.2 Transducer - The transducer shall have four load-factor levels and a reset level. The levels shall be the nominal values specified in Table 1.

TABLE I

Nominal Load-Factor and Reset Levels

MS Part No.	Counter 1	Counter 2	Counter 3	Counter 4	Reset Levels
MS25447-1	2.0	2.5	3.0	3.5	1.5 + 0 - 0.25
MS25447-2	2.5	3.5	4.0	4.5	1.75 ± 0.2
MS25447-3	2.5	3.5	4.5	5.5	1.75 ± 0.2
MS25447-4	3.0	4.0	5.0	6.0	2.0 ± 0.2
MS25447-5	4.0	4.5	5.0	5.5	2.5 ± 0.2
MS25447-6	4.0	5.0	6.0	7.0	2.5 ± 0.2
MS25447-7	5.0	6.0	7.0	8.0	3.0 ± 0.2
MS25447-8	5.5	6.0	6.5	7.0	3.25 ± 0.2
MS25447-9	6.0	7.0	8.0	9.0	3.5 ± 0.2
MS25447-10	6.0	7.0	8.5	10.0	3.5 ± 0.2

3.4.2.1 Size - The maximum outline dimensions of the transducer shall be as shown in MS25447.

3.4.2.2 Weight - The weight of the transducer shall not exceed 1.5 pounds.

3.4.2.3 Markings - Each transducer shall have clearly indicated on the case the location of the center of gravity of the seismic mass and the direction of mounting to obtain positive vertical accelerations. Each transducer shall be marked, if applicable, as follows:

Installed transducer and indicator shall be of the same manufacturer.

3.4.3 Indicator - The indicator shall contain four counters, each counter displaying four digits. The design shall be in accordance with MS25448. The four counters shall not respond to sinusoidal or square wave input voltages at frequencies of 4.0 cps and above. The circuit of the indicator shall prevent the counters from counting until the load-factor level has returned to the reset level or below, as specified in Table 1.

3.4.3.1 Size - The size of the indicator unit shall be in accordance with MS25448.

3.4.3.2 Weight - The weight of the indicator unit shall not exceed 3.5 pounds.

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3.4.3.3 Window Material - The window material shall be clear and free from flaws which would interfere with normal readings of the counters and shall otherwise conform to this specification.

3.4.3.3.1 Distance from Window to Counter - The distance from any window to its counter shall be as small as practicable and shall not exceed 0.156 inch.

3.4.3.4 Counters - Four counter units shall be provided. The counters shall be viewed through windows in the indicator case and shall indicate the occurrence of load factors.

3.4.3.5 Counter Markings - All the counter dials shall have the following numerals: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. The digit height shall be 0.200 inch + 0.01 inch or - 0.03 inch, and the line thickness shall be 0.030 inch ± 0.001 inch. The style and proportions of the numerals placed on the dial shall conform to MS33558. The colors shall be lusterless white color no. 37875 and lusterless black color no. 37038 of FED-STD-595.

3.4.3.5.1 Visibility of Counter Numerals - The numerals on the counter shall be visible from any point within the frustrum of a cone whose sides make an angle of 30° with a perpendicular to the counter and whose small diameter is the aperture of the indicator case. When the indicator is viewed from all angles forward of the plane of the indicator face, only the numerals representing the present count shall be visible; the next advancing and next receding numerals shall not be visible.

3.4.3.6 Markings - Each indicator shall be marked, if applicable, as follows:

Installed indicator and transducer shall be of the same manufacturer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall also include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181. Each sample shall meet the requirements of 4.2.2.1 and 4.2.2.2.

4.2.2.1 Acceleration Test - With electrical power off, the accelerometer group shall not record any counts when subjected to 10g along each axis for 1 minute. Upon completion of this test, the equipment shall be subjected to the static calibration test.

4.2.2.2 Impact Test - With electrical power off, the indicator shall not record any counts when subjected to three impacts of 25g in each direction along each of its three mutually perpendicular axes for a total of 18 impacts. Each impact shall be a half sine-wave with a rise time of approximately 5.5 milliseconds and a time duration of 11±1 millisecond.

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4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitue cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause of deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance shall be by lots, and acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum each equipment accepted shall have passed the following tests, conducted in the order listed:

Examination of Product	4.4.1.1
Operational Test	4.4.1.2
Dielectric Test	4.4.1.3
Static Calibration Test	4.4.1.4
Leak Test	4.4.1.5
Indicator Circuit Test	4.4.1.6
Indicator Counting Rate Test	4.4.1.7

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Dielectric Test -

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4.4.1.3.1 Transducer - A potential of 200 volts DC shall be applied between isolated pins and between pins and case for a period of 5 seconds. During application of this voltage, the insulation resistance as measured by a suitable "megger" shall be not less than 3.5 megohms.

4.4.1.3.2 Indicator - A potential of 500 volts DC shall be applied between isolated pins and between pins and case for a period of 5 seconds. During application of this voltage, the insulation resistance as measured by a suitable "megger" shall be not less than 3.5 megohms.

4.4.1.4 Static Calibration Test - A static calibration shall be conducted on each transducer connected to an indicator to determine that the load-factor and reset levels are within the limits of Table 1. All calibrations shall remain within ± 0.1g of the nominal levels specified in Table 1 for load-factors of 2.0 to 7.0g and ± 0.15g for load-factor levels of 7.5 to 10.0g. These limits shall apply after the transducer is subjected to any or all environmental tests. An electrical vibrator set at 60 cps with a maximum amplitude of 0.002 inch when attached to the transducer shall be used to impart vibrations to the transducer. All calibratinns shall remain within the limits specified by this paragraph when the calibrations are obtained with and without the vibrator. The reset channel shall be calibrated for increasing and decreasing load-factor. The difference between the calibration levels of the reset channel under increasing and decreasing load-factors shall remain within the limits specified by this paragraph. For Preproduction, Initial Production, Acceptance, Reliability Assurance, and Special Tests the Static Calibration Test shall be performed on an equal number of transducers and indicators connected together as a counting accelerometer group. The transducers and indicators shall be randomly interchanged on subsequent tests performed following each environmental test.

4.4.1.5 Leak Test - Each transducer shall be tested for case leakage with either a mass spectrometer or by an equally sensitive leak test. The gas, or mixture of gases, in the case, adjusted to a pressure differential of one atmosphere, shall not have a leakage rate from the case exceeding 1 micron cubic foot per hour (10^{-5} cubic centimeters per second). Oil-filled transducers shall be leak tested, prior to filling with oil, by means of the mass spectrometer. The measured leak rate shall not exceed an equivalent of 1 cubic centimeter of air/year/inch of seal.

4.4.1.6 Indicator Circuit Test - The counter shall not count when subjected to the following test: Three continuous load cycles that vary from above the highest load-factor level to above the reset level, but lower than the first load level. NOTE: The counters shall have counted once when returned to a value less than the reset level after completion of the test.) The first counter shall not count when subjected to the following test: Twelve continuous load cycles, at a minimum rate of 1 cps, varying from just below the first load-factor level to just below the reset level.

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4.4.1.7 Indicator Counting Rate Test - The indicator counters shall be subjected to a sinusoidal or square wave input of 1, 2 and 3 cps. The counters shall operate for a minimum of 150 counts and each counter shall be advanced by the same number of incremental counts. There shall be no counts at 4 cps and above.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the Individual Tests. Counter Accelerometer Groups shall be selected at random for Sampling Tests by the Government Inspector in accordance with the following:

Quantity of Counting Accelerometer Groups Offered for Acceptance	Quantity to be Selected for Testing
100 or less	5
Each additional 100 or fraction thereof	5

4.4.2.1 Scope of Tests - Each sample selected shall be subjected to the following tests, conducted in the order listed:

Vibration Test	4.4.2.2
Temperature Shock Test	4.4.2.3
Dynamic Tests	4.4.2.4
High Temperature Test	4.4.2.5
Low Temperature Test	4.4.2.6

4.4.2.2 Vibration - Tests shall be conducted under ambient room conditions and also at -54°C and 125°C. The time specified in Table 3 shall be divided as follows: 50% at room ambient, 25% at -54°C, and 25% at 125°C. Total time of the tests shall be 9 hours for the three mutually perpendicular axes of the counting accelerometer group.

TABLE 3

Vibration Test Schedule

(Time shown refers to one axis of vibration)

Number of Resonances	0	1	2	3
Total Vibration Time at Resonance*	-	½	1 hr	1½ hrs
Cycling Time	3 hrs	2½ hrs	2 hrs	1½ hrs

* 30 minutes at each resonance

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The counting accelerometer group shall be normally mounted on the vibration equipment. The mounting shall simulate service installation orientation. The test fixture shall be free of resonances through 1000 cps. Power shall be applied to the counting accelerometer group during the following steps:

Step 1, Resonance Survey - Resonance modes and/or minimum performance points of the equipment shall be determined by varying the frequency of applied vibration slowly through the range of 5 to 500 cps at a double amplitude not exceeding those indicated in Figure 2. Individual resonance survey shall be conducted with vibration applied along each of the three mutually perpendicular axes of the counting accelerometer group.

Step 2, Resonance Vibration - With the counting accelerometer group mounted on the test fixture, vibration shall be applied at each of the individual resonance points obtained in step 1 at the double amplitude indicated in Figure 2. Vibration shall be 30 minutes at each resonance and/or minimum performance points. If more than 3 resonant modes and/or minimum performance points are noted for any one axis, the three most severe shall be used for each axis. At the completion of this step, the counting accelerometer group shall be inspected for evidence of any mechanical failure. Any counting accelerometer counts recorded during the vibration tests shall constitute a failure.

Step 3, Cycling - The counting accelerometer group shall be vibrated in accordance with the requirements of Figure 2 over the frequency range 5 to 500 cps. The variation in frequency shall be at a logarithmic rate; however, if logarithmic cycling is not available, a linear rate of frequency change may be used. The cycle from 5 to 500 cps and return to 5 cps shall be accomplished in approximately 15 minutes including at least 3/4 minute in the region below 25 cps if the linear rate of frequency change is used. The Vibration Test Schedule, Table 3, indicates the times required for cycling. At the completion of this step, the counting accelerometer group shall be inspected for evidence of any mechanical failure. Any accelerometer counts recorded during this test shall constitute a failure.

Steps 1 through 3, above, shall be repeated for each of three mutually perpendicular axes of the counting accelerometer group.

4.4.2.3 Temperature Shock Test - The counting accelerometer group shall be subjected to the temperature shock test of MIL-E-5272, Procedure I, except that the high temperature shall be 125°C and low temperature shall be -54°C. Upon completion of the above test the accelerometer group shall be subjected to the static calibration test.

4.4.2.4 Dynamic Test - Counting accelerometer groups shall be subjected to the dynamic tests as specified herein. The transducers and indicators shall be randomly interchanged on subsequent tests following each environmental test.

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4.4.2.4.1 Dynamic Calibration - The transducer shall be calibrated against known sinusoidal acceleration inputs of 1, 2, and 2.5 cps. The counting accelerometer group shall count at the single-amplitude, nominal load-factor levels prescribed in Table 1, within the limits specified in 3.4.1.6. The calibration shall remain within these specified limits after the counting accelerometer group has been subjected to any or all of the environmental tests.

4.4.2.4.2 Calibration with Superimposed Vibration - The test shall be conducted with the transducer and indicator connected electrically as a group. The transducer shall be subjected to static load-factor levels upon which are superimposed sinusoidal inputs along the sensitive axis of the transducer. The superimposed vibration shall be applied logarithmically or linearly at a rate over the range of 5 to 120 cps at the amplitudes specified in Table 2 for a duration of 5 minutes. The following sequence shall apply:

(1) Apply a static load-factor level to the transducer until the reset level as determined by the static calibration (4.6.3) is reached. Apply the superimposed vibration as specified above. There shall be no counts nor storage of counts.

(2) Increase the static load-factor level to a level 0.20g below the nominal value of the first load-factor level of Table 1. Apply the superimposed vibration as specified above. There shall be no counts nor storage of counts.

(3) Increase the static load-factor level in increments of 0.05 g and apply the superimposed vibration as specified above, at each increment until the load level with vibration is determined. This load level shall be within ± 0.15g of the nominal load-factor level values in Table 1 for the specified range of the transducer.

(4) Repeat (2) and (3) for each of the next three load-factor levels.

(5) Decrease the static load-factor level until the reset level of the transducer is reached. Apply the superimposed vibration as specified above. There shall be only one count registered on the indicator for each of the four load-factor levels. Figure 3 is a typical example of the superimposed vibration.

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TABLE 2

Frequency (cps)	Amplitude (inch or "g")
5-40	0.10 inch - double amplitude
41-120	3.5g - single amplitude

4.4.2.5 Low Temperature Test - The accelerometer group shall be subjected to a temperature of -54°C for a period of 8 hours. At the end of the 8 hours and while still at -54°C, the indicator counting rate test, the indicator circuit test, the static calibration test, and the dynamic tests shall be performed.

4.4.2.6 High Temperature Test - The accelerometer group shall be subjected to a temperature of 125°C for a period of 4 hours. At the end of 4 hours, the group shall be removed from the temperature chamber and within 10 minutes thereafter be subjected to the indicator counting rate test, the indicator circuit test, the static calibration test, and the dynamic tests.

4..4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. For the applicable test plan 50,000 operations shall be considered one (1) "MTBF". An operation shall be the cycling of the transducer from zero load-factor level to maximum load-factor level and return to below the reset load-factor level. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted.

4.4.3.1 Qualification Phase - Prior to acceptance of equipments under the contract or order, a minimum of three (3) equipments (transducers and indicators) shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The type designation and the quantities of the equipment together with assigned load-factor levels for the tests shall be designated by the procuring activity. The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan IV shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the section entitled "Sampling Phase of Product on Reliability Tests". For the Sampling Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan VI shall be used to determine the length of the tests until the accept-or-reject decision is reached. The type designation and the quantities of the equipment together with assigned load-factor levels for the tests shall be designated by the Government Inspector.

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4.4.3.2.1 Lot Size for Sampling Phase - The equipments constructed during one month shall be one lot. Unless otherwise specified in the contract or order, no equipments shall be shipped until an Accept decision is reached under each lot.

4.4.3.3 Test Details - The test details such as the length of the Test Cycle, the temperature cycle (4.4.3.3.1), the performance characteristics to be measured (4.4.3.4), special failure criteria (4.4.3.5), preventive maintenance to be allowed during the test (4.4.3.6), etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.3.3.1 Temperature Cycle - Each accelerometer group undergoing test shall be subjected to a minimum of three (3) complete temperature cycles as required by the specified test level. At least once during the high temperature cycle, at the stabilization portion of each temperature cycle, the dynamic test shall be performed.

4.4.3.4 Performance Characteristics to be Measured - The performance characteristics to be measured shall be as specified in 6.9.

4.4.3.5 Failure Criteria - In addition to the requirements of MIL-STD-781, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the acceptance requirement 6.9 at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

(2) Whenever a complete operation of the counting accelerometer group does not count on the appropriate indicator counter, it is a failure. Under the same condition, an excessive number of counts is a failure.

4.4.3.6 Preventive Maintenance - During the period of the tests no preventive maintenance may be performed upon the equipment.

4.4.4. Special Tests - Special Tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

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4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an approved engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This shall be authorized by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.5 Life Test - The contractor shall furnish all life test samples and shall be responsible for accomplishing the life test. The equipment shall have a total operating life of 100,000 operations with reasonable servicing and replacement of parts. A 100,000 operation life test shall be conducted on the transducer and indicator. An operation shall be the cycling of the transducer from zero load-factor level to maximum load-factor level and return to below the reset load-factor level. All four counters shall count once for each operation. The life test for the equipment shall be conducted on those samples representative of the equipment being produced. The equipment shall have all four load-factor levels and the reset level operated in the following sequence:

10,000 operations at -54°C
10,000 operations at +125°C
10,000 operations at room temperature
10,000 operations at -54°C
10,000 operations at +125°C
10,000 operations at room temperature
10,000 operations at -54°C
10,000 operations at +125°C
20,000 operations at room temperature

Upon completion of the above tests, the equipment shall be subjected to and meet the requirements of 4.4.1.4 and 4.4.2.4.

4.5.1 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.2 Performance Check - At every 10,000 operations intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

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4.5.3 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of operations prior to failure

4.5.4 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Equipment Failure - Should a failure occur during the Individual, the Sampling, Reliability Assurance, Special or Life Tests the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a new test in the individual test section to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.7 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation

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of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.8 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.9 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.10 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - The counting accelerometer groups covered by this specification are intended for use on aircraft to indicate the number of times that the aircraft have experienced selected, predetermined, acceleration load-factor levels during airborne operations.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltage of 27.5±0.5VDC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

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6.3 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (See 5.1).

(3) Type designation (MS part number), the quantity, and the load-factor levels for the Reliability Qualification Phase Tests.

(4) The designated model, drawings and other technical data to be made available by the procuring activity.

6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

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6.6 Non-Repairable Subassemblies - As a general rule, non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 200,000 operations, and for many applications this figure must be nearer 600,000 operations.

6.7 Type Designations - The parentheses (*) when used in the equipment type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.8 Designation - The transducer shall be designated according to load-factor levels. Indicators shall be used with transducers of any load-factor levels.

6.9 Performance Characteristics to be Measured - The transducers when subjected to nominal load-factor levels shown in Table 1 within the limits specified in 4.4.1.4 and 4.4.2.4, shall generate signal outputs to the indicator. The indicator shall receive these signals and display on its four counters the exact number of times this has occurred. This transducer output signal generation together with the related indication signal receipt and exact counter display for a nominal load-factor level, as described herein, shall constitute an operation.

6.10 Definitions -

6.10.1 "g" - The acceleration due to gravity.

6.10.2 Load-Factor Level - The load-factor level is equivalent to absolute acceleration in g-units.

6.10.3 Count - The entire action performed by a Counting Accelerometer Group to satisfactorily complete registry of one operation.

6.10.4 Insulation Breakdown - Insulation breakdown shall be defined as an alternating current leakage impedance of less than 200,000 ohms, or a leakage current of 2.5 milliamperes.

6.10.5 Amplitude - Whenever the words "double amplitude" are specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

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6.10.6 Hermetic Seal - A hermetic seal shall be a perfectly closed and airtight seal made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals accomplished by gaskets.

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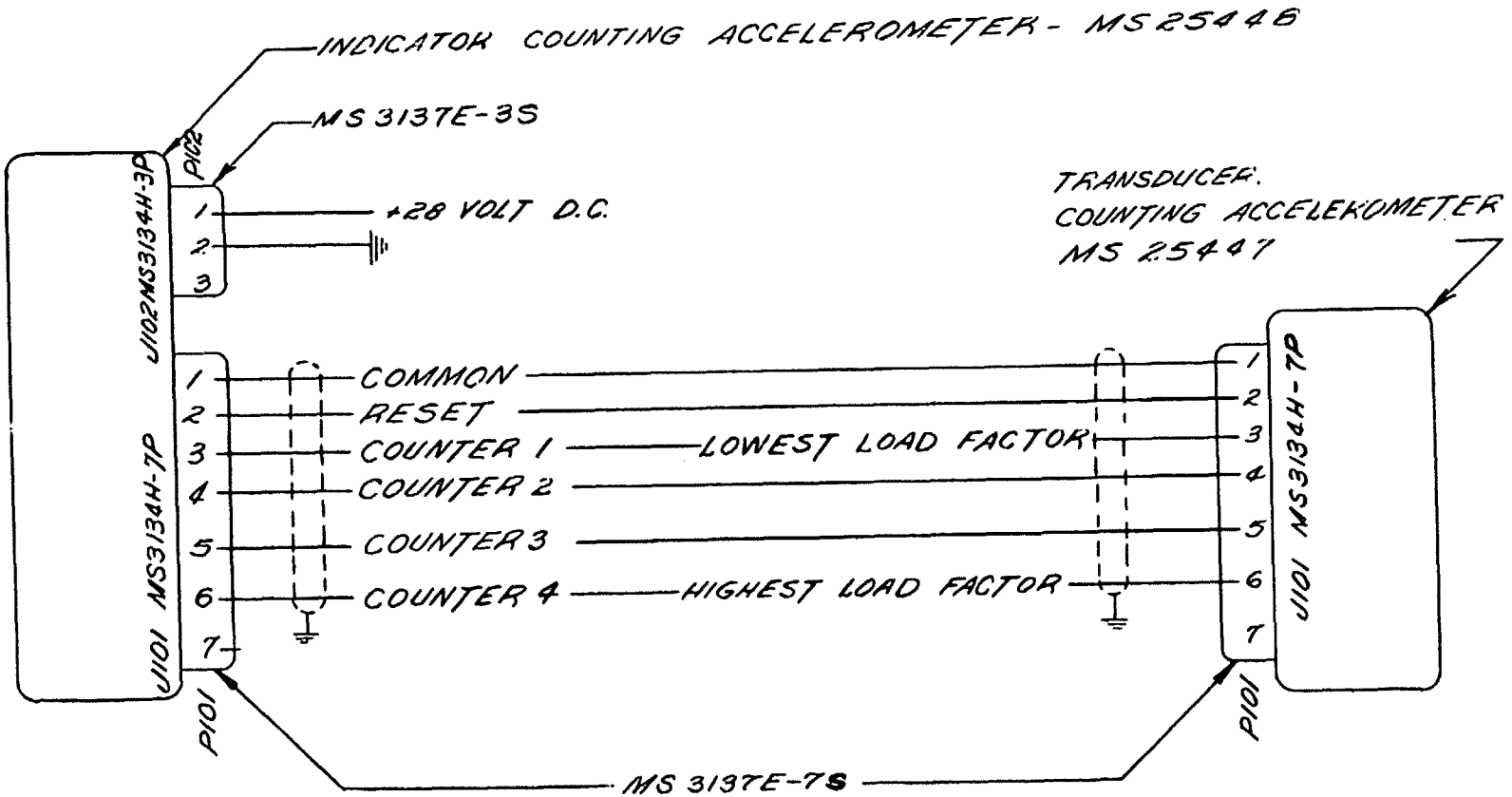
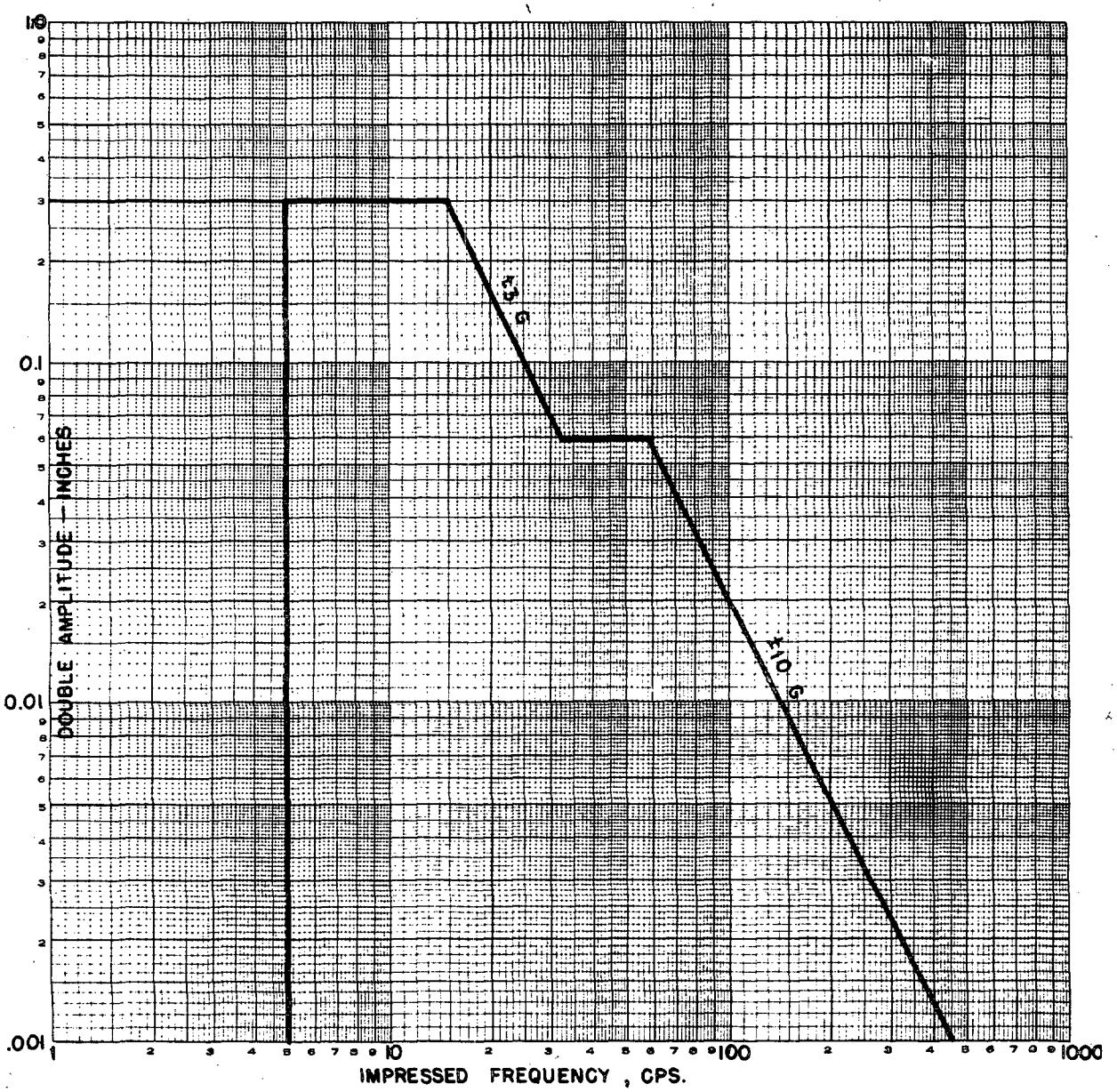


FIGURE 1

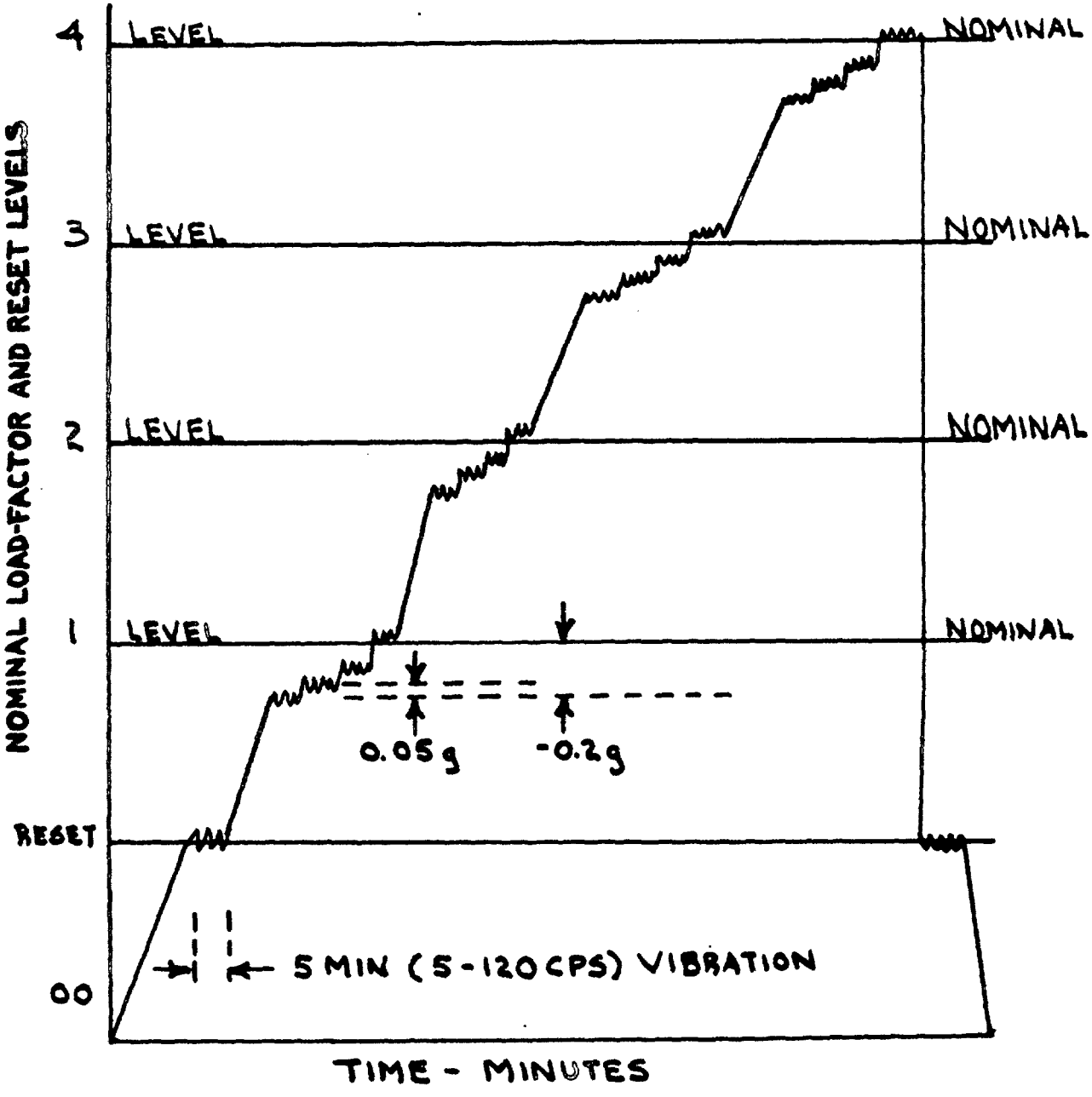
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VIBRATION REQUIREMENTS.

FIGURE 2

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TYPICAL EXAMPLE OF SUPERIMPOSED VIBRATION APPLIED FOR A TOTAL OF 90 MINUTES AT THE RESPECTIVE LOAD LEVELS

FIGURE 3

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-A-22145B(AS) ACCELEROMETER GROUP, COUNTING TYPE DESIGNATION MS25447 and MS25448

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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